Query: why do people use the NASA-TLX?

Support note: every page this answer draws on is `status: draft` — the NASA-TLX cluster was ingested today (Hart1988 at 14:32, Hart2006 at 15:48) and has not yet been through `/audit`. So the whole answer is unverified draft coverage, marked `*[from draft]*` throughout. It is wiki-grounded only: claims rest on the draft source/concept pages and their own locators; the raw PDFs were not reopened this run (routine question). This closes the gap named by the 13:48 zero-coverage query — the wiki is no longer silent on workload measurement.

Short answer

People use the NASA-TLX because it is a fast, validated, multidimensional way to measure a quantity — a human operator's mental workload — that has no objective gauge, and because it gives a diagnostic six-part profile (not just one number) while having become the field's de-facto standard, so results are comparable and the tool is trusted. The reasons, strongest first:

- It measures something with no objective ruler, so self-report is the right tool. Mental workload is the cost an operator pays to reach a level of performance; it has no physical units, no agreed zero or maximum, and no external ground truth, so the operator's own felt experience is the quantity of interest and subjective self-report is the direct measure of it. NASA-TLX is a refined, standardized instance of that paradigm (Hart1988DevelopmentON, Introduction / Information Provided by Subjective Ratings, pp. 139, 141). *[from draft]*
- It is diagnostic — it shows where the load comes from. Instead of one global number it keeps six subscales (Mental, Physical, and Temporal Demand, Performance, Effort, Frustration), so a high score can be traced to its source (e.g. temporal vs mental demand) and acted on rather than just noted. This is the headline reason and the one that survives 20 years of use: analyzing the subscales individually is singled out as a continuing strength (NASA-TLX Why It Matters; Hart1988DevelopmentON, pp. 161, 174; Hart2006NasaTaskLI, Methodological Issues, p. 4). *[from draft]*
- It is more reliable across people than a single global rating. Weighting each rater's six components by their own task-specific pairwise importance judgements lowers between-subject variability versus one overall rating — about 20% on average and up to 46% — at no loss of sensitivity, so the score is more consistent person-to-person (Hart1988DevelopmentON, p. 152). *[from draft]*
- It is fast, portable, and stable. The six ratings take under a minute and the weights under two; paper, computer, and verbal forms correlate 0.94–0.96; test–retest reliability is 0.83 at four weeks; and it is easier to administer than SWAT's conjoint card-sort (NASA-TLX Why It Matters; Hart1988DevelopmentON, pp. 175–176). *[from draft]*

- It is the de-facto standard, so people use it partly because everyone else does. Twenty years on it is translated into more than a dozen languages, described in textbooks and taught in courses, used as the benchmark other measures are judged against, and applied far beyond its aviation origin (driving, medicine, computing, teleoperation). That standardization makes results comparable across studies and lowers the cost of choosing a measure (Hart2006NasaTaskLI, Background / Summary, pp. 1, 4). *[from draft]*

The honest tension worth knowing

The pairwise weighting — the 1988 design's signature contribution and one of the reasons above — is the part people most often drop. Raw TLX (RTLX), which skips the weighting and just averages or sums the six ratings, is the single most common modification, and across the 29 studies that compared it to the weighted score it was more, less, or equally sensitive — "you can take your pick" (Hart2006NasaTaskLI, Methodological Issues, p. 3). The 1988 paper itself flags that the original a-priori weights "did not achieve the desired level of improvement in statistical sensitivity"; it is the per-task weighting that delivers the reliability gain (Hart1988DevelopmentON, p. 166). *[from draft]*

So in practice the durable reasons people reach for NASA-TLX are the diagnostic six-subscale profile, the speed, and the standardization — more than the weighting step its name is built around. The wiki records this as disconfirming evidence on NASA-TLX, not as a contradiction (Hart2006 qualifies the weighting's payoff; it does not dispute the 1988 between-subject-variability finding, which is a different metric). *[from draft]*

What the wiki cannot yet tell you

- The numbers above (20% variability reduction, 0.83 test–retest, 0.94–0.96 cross-form correlations, the 29-study RTLX wash) are wiki-grounded from draft pages, not re-checked against the raw this run. They were raw-verified at ingest today per the ingest reports, but `/audit` has not independently confirmed them. For any high-stakes use, reopen the Hart papers or run `/audit` first.
- The wiki has no teaming-workload angle yet: workload here is the single-operator human-factors construct (Hart's aviation lineage). How a human's workload is measured specifically when teaming with an AI — the bridge to the human-autonomy team thread and your candidacy's FM-MAS focus — is not covered. The connection is asserted as adjacency only.
- Raw TLX is single-source (Hart2006 only), so the "weighting is a wash" evidence rests on one retrospective by the instrument's own creator.

Recommended ingests

- Byers, J. C., Bittner, A. C., & Hill, S. G. (1989) — "Traditional and Raw Task Load Index (TLX) Correlations: Are Paired Comparisons Necessary?" (in Advances in Industrial Ergonomics and Safety I) — the RTLX primary; would de-single-source Raw TLX and ground the weighting-vs-raw question on something other than Hart's own retrospective.
- Rubio, S., Díaz, E., Martín, J., & Puente, J. M. (2004) — "Evaluation of Subjective Mental Workload: A Comparison of SWAT, NASA-TLX, and Workload Profile Methods" (Applied Psychology) — a head-to-head comparison; would de-single-source subjective workload measurement and give SWAT and the Workload Profile their own grounding instead of NASA-TLX-internal mentions.
- Cooper, G. E., & Harper, R. P. (1969) — "The Use of Pilot Rating in the Evaluation of Aircraft Handling Qualities" (NASA TN D-5153) — the Cooper-Harper scale, named as a sibling instrument on subjective workload measurement with no page or source behind it.

Process note (not an ingest): the four NASA-TLX pages and two Hart sources are all `status: draft`. The highest-value next step for this topic is `/audit` (or `/checkup`) to fact-check the cluster against the raws and lift it to verified — then a later query can answer this without the draft caveat.